

PROJECT REPORT

SMARTLEAD

AI-Powered Digital Service Discovery & Lead Management System

A Multi-Domain Collaborative Student Project

Digital Marketing • UI/UX • HTML/Tailwind • React • Python/Django • PostgreSQL • Machine Learning

1. Executive Summary

SmartLead is a full-stack web platform designed for an organization that provides digital and technology-related services. The platform allows visitors and registered users to discover services, search and filter service offerings, view service details, and submit enquiries. Each enquiry can become a business lead that is managed by the organization's staff or administrators.

The system is not intended to be a conventional e-commerce application. Its central business flow is Service Discovery → Enquiry → Lead → Follow-up → Conversion. A supervised machine-learning component will analyze historical lead data and predict the probability that a new lead will convert into a customer.

The project is deliberately structured as a collaborative, multi-domain development exercise. Digital Marketing students, UI/UX students, HTML/Tailwind students, React students, Python/Django students and Machine Learning students will contribute to one integrated product.

2. What the Project Represents

SmartLead represents a digital service company's customer-acquisition and lead-management platform.

1. A company publishes its services on the platform.
2. A visitor discovers a service through the website, search or a marketing campaign.
3. The visitor reads the service information and submits an enquiry.
4. The enquiry is stored and managed as a lead.
5. An administrator or staff member follows up with the lead and updates its status.
6. The machine-learning model predicts the probability of conversion using historical and current lead attributes.
7. The organization can use dashboards and analytics to understand service interest, lead sources and conversion activity.

Core business flow:

Service → Enquiry → Lead → Follow-up → Conversion

3. Difference from an E-Commerce Application

SmartLead should not be treated as a shopping-cart or online-ordering system. The platform focuses on services and customer enquiries rather than physical product transactions.

Typical E-Commerce	SmartLead	Primary Purpose
Product → Cart → Checkout → Payment → Order	Service → Enquiry → Lead → Follow-up → Conversion	Customer acquisition
Physical/digital product transaction	Service discovery and enquiry	Generate and manage prospects
Order management	Lead and follow-up management	Track business opportunities
Payment and delivery	No mandatory payment/shipping flow	Focus on service enquiries

4. Project Objectives

- Develop a complete service-discovery web platform for an organization.
- Provide a public/general-user interface for discovering services and submitting enquiries.
- Provide an administrative interface for managing users, services, enquiries and leads.
- Implement REST APIs using Django REST Framework for frontend-backend communication.

- Use PostgreSQL as the primary database.
- Implement role-based access and authentication.
- Introduce supervised machine learning for lead-conversion prediction.
- Give Digital Marketing students a practical role in keyword research, SEO and marketing analytics.
- Create a real collaborative workflow across UI/UX, frontend, backend and ML teams.

5. Major Functional Modules

5.1 Service Discovery Module

- Browse available services.
- Search services by keyword.
- Filter and sort services.
- View service details, features and packages.
- View categories and related services.
- Mark services as favourites/wishlist.

5.2 User Management Module

- User registration and login.
- Logout and password reset/change.
- Profile management.
- View enquiry history.
- View enquiry status and notifications.

5.3 Enquiry Management Module

- Submit an enquiry for a selected service.
- Capture requirement, contact details, source and other relevant information.
- Generate a lead from an enquiry.
- Allow users to track the status of their enquiries.

5.4 Lead Management Module

- View and search leads.
- Filter leads by status, service, source and date.
- Assign leads to staff/admin users.
- Update lead status.
- Record follow-up notes.
- Schedule follow-ups.
- Maintain lead history.
- Mark leads as converted or not converted.

5.5 AI-Based Lead Prediction Module

- Train a supervised classification model using historical lead data.
- Use Logistic Regression as the initial model.
- Predict the probability of lead conversion.
- Display the prediction to authorized administrative users.
- Classify leads into practical probability bands such as high, medium and low.

5.6 Admin & Business Management Module

- Manage users and roles.
- Manage services and categories.
- Manage enquiries and leads.
- Manage reviews/feedback.
- View business analytics.
- Manage relevant SEO/content information.

5.7 Review & Feedback Module

- Allow users to provide ratings and feedback.
- Allow administrators to moderate reviews.
- Use feedback data for business reporting where appropriate.

5.8 Notification Module

- Notify users about enquiry status changes.
- Notify administrators about new enquiries.
- Highlight follow-ups that are due.
- Highlight high-probability leads.

6. General User Journey

8. User visits the platform.
9. User searches or browses available services.
10. User opens a service details page.
11. User registers/logs in if required.
12. User submits an enquiry describing the requirement.
13. The system stores the enquiry and creates a lead.
14. The lead becomes visible to authorized administrative users.
15. The system can generate a machine-learning conversion prediction.
16. The administrator follows up with the lead and updates its status.
17. The lead may eventually be marked as converted or not converted.

7. Administrative Journey

- Admin logs into the administrative interface.
- Admin views dashboard statistics.
- Admin manages services and categories.
- Admin reviews new enquiries.
- Admin assigns and manages leads.
- Admin records follow-ups and changes lead status.
- Admin views ML-based conversion predictions.
- Admin reviews analytics related to services, traffic sources, enquiries and conversions.

8. Machine Learning Component

The machine-learning component is intended to give the Python/ML students a meaningful role in the business workflow. The initial supervised learning problem is binary classification: predict whether a lead is likely to convert.

8.1 Proposed Algorithm

Logistic Regression — a supervised classification algorithm suitable for an initial, explainable implementation.

8.2 Example Input Features

Feature	Example
Traffic/lead source	Google
Pages viewed	10
Previous enquiries	3
Time on website	500 seconds
Service selected	Web Development
Budget range	₹50,000
Engagement score	85

8.3 Target Variable

Converted: 1 = Yes, 0 = No

8.4 Example Prediction

For a new lead, the model may produce a conversion probability such as 81%. The administrative dashboard can display this as a high conversion-probability lead. The displayed probability is a model output, not a guarantee of conversion.

9. Digital Marketing & SEO Component

- Conduct keyword research for important service pages.
- Identify primary, secondary and long-tail keywords.
- Study search intent and keyword competitiveness.
- Prepare SEO titles and meta descriptions.
- Recommend H1/H2 structures and URL patterns.
- Define image alt-text requirements and internal-linking opportunities.
- Develop content recommendations for service landing pages.
- Track traffic sources, campaign sources, landing pages, enquiries and conversion activity where analytics data is available.

Example keyword structure:

Primary: web development services

Secondary: React development, Django development, Python development

Long-tail: affordable Django development services

10. UI/UX Component

UI/UX students will create the initial product experience in Figma for two major interfaces.

10.1 General User Interface

- Home page
- Service listing
- Service details
- Login/Register
- User dashboard
- Enquiry form
- Enquiry history
- Profile
- Wishlist/Favourites

10.2 Admin Interface

- Admin login
- Dashboard
- User management
- Service management
- Category management
- Lead management
- Follow-up management
- Analytics
- ML prediction view
- Settings

10.3 Design Expectations

- Desktop design
- Tablet/mobile responsive considerations
- Navigation
- Forms
- Cards
- Tables
- Dashboard components
- Buttons and modals
- Error states
- Empty states

11. HTML + Tailwind CSS Component

The HTML/Tailwind team will take the approved Figma designs and convert them into responsive frontend templates. The focus is visual implementation and responsive behavior rather than complete business logic.

- HTML5 page structure
- Tailwind CSS responsive layouts
- Reusable UI sections
- Forms and tables
- Responsive navigation
- Mobile/tablet/desktop layouts
- Handoff of templates to the React team

12. React Frontend Component

- Convert responsive templates into a functional React application.
- Implement routing.
- Create reusable components.
- Implement form handling and client-side validation.
- Consume Django REST APIs using Axios or Fetch.
- Implement authenticated user flows.
- Create separate user and admin application areas.
- Display service, enquiry, lead and analytics data received from APIs.

Illustrative routes:

/services | /services/:id | /login | /register | /profile | /enquiries | /admin/dashboard | /admin/leads

13. Python/Django Backend Component

- Develop backend business logic using Python and Django.
- Use Django REST Framework to create APIs.
- Implement authentication and authorization.
- Implement role-based permissions.
- Create database models and relationships.
- Integrate PostgreSQL.
- Manage enquiries, leads, follow-ups, services, users and reviews.
- Provide ML prediction API integration.
- Configure and customize the Django Admin panel.

13.1 Possible Django Apps

accounts • services • categories • enquiries • leads • reviews • analytics • ml_prediction

14. Indicative REST API Endpoints

Method	Endpoint	Purpose
POST	/api/auth/register/	Register user
POST	/api/auth/login/	Authenticate user
POST	/api/auth/logout/	Logout
GET	/api/services/	List services
GET	/api/services/{id}/	Service details
POST	/api/enquiries/	Submit enquiry
GET	/api/leads/	List/manage leads
GET	/api/leads/{id}/	Lead details
POST	/api/leads/{id}/followup/	Record follow-up
POST	/api/ml/predict/	Generate conversion prediction

15. Database & Core Entities

PostgreSQL will be used as the primary relational database.

Entity	Purpose
User	Authentication, profile and roles

Category	Service categorization
Service	Published service information
ServiceImage	Service media
Wishlist	User favourite services
Enquiry	User service enquiry
Lead	Business lead created from enquiry
LeadFollowUp	Follow-up history
Review	User rating and feedback
SEOContent	SEO-related page/content information
MLPrediction	Prediction output and related model metadata

16. Role-Based Access Control

Role	Typical Access	Responsibilities
General User	Own profile, services, enquiries, wishlist, reviews	Discover services and submit/manage own enquiries
Staff/Admin	Operational modules and leads	Manage services, enquiries, leads and follow-ups
Super Admin	Full administrative access	Manage users, roles, permissions and system-wide configuration

17. High-Level Technology Architecture

Digital Marketing → UI/UX/Figma → HTML/Tailwind → React → Django REST API → PostgreSQL, with Python/Scikit-learn providing the ML prediction component and Django Admin providing internal administration.

- Digital Marketing: keywords, SEO and content strategy
- UI/UX: Figma designs for User and Admin interfaces
- HTML + Tailwind: responsive templates
- React: functional frontend and API integration
- Django + DRF: backend, authentication and REST APIs
- PostgreSQL: persistent relational data
- Python + Scikit-learn: supervised ML model
- Django Admin: internal administration

18. Department-Wise Responsibilities

Department	Technology/Tools	Primary Deliverables
Digital Marketing	SEO/analytics tools	Keyword research, SEO plan, content strategy, campaign/traffic analysis
UI/UX	Figma	User and Admin designs, responsive design specifications
HTML/Tailwind	HTML5, Tailwind CSS	Responsive templates and reusable visual components
React	React, Axios/Fetch	User/Admin frontend, routing, forms and API integration
Python/Django	Python, Django, DRF	Backend, business logic, authentication, REST APIs and Django Admin
Database	PostgreSQL	Database schema, relationships and data management
Machine Learning	Python, Scikit-learn	Lead conversion classification model and prediction service
Testing/QA	Manual/API testing	Functional, UI and API validation

19. End-to-End Collaborative Workflow

18. Digital Marketing identifies target keywords, search intent and content requirements.
19. UI/UX creates the approved Figma experience for the User and Admin portals.
20. HTML/Tailwind converts the approved designs into responsive templates.
21. React converts the templates into a functional frontend application.
22. Django develops the backend, database models, authentication and REST APIs.
23. React integrates with the Django REST APIs.
24. Historical lead data is prepared for the supervised ML model.
25. The ML team trains and evaluates the Logistic Regression model.
26. The prediction service is integrated into the lead-management workflow.
27. Testing is performed across UI, API, backend and ML functionality.
28. The integrated application can then be prepared for deployment.

20. Initial Project Scope

The first implementation should concentrate on the following core scope:

- Service discovery and service details.
- User registration/login and profile.
- Enquiry submission and enquiry tracking.
- Lead management and follow-ups.
- Admin dashboard.
- Role-based access.
- SEO/content requirements.
- REST API integration.
- PostgreSQL database.
- Supervised ML lead-conversion prediction using Logistic Regression.
- Basic analytics and reporting.

Features such as online payment, shopping cart, shipping, inventory and order fulfilment are outside the initial scope unless later approved.

21. Expected Outcome

The final outcome will be one integrated web application rather than a collection of unrelated departmental assignments. Each student group will produce a deliverable that is consumed by another group, creating a realistic software-development workflow.

The project will demonstrate how digital marketing, product design, frontend engineering, backend engineering, database development and machine learning can work together to solve a single business problem.

23. Proposed Final Project Definition

SmartLead is an AI-powered digital service discovery and lead management platform that enables users to discover services and submit enquiries, while organizations manage those enquiries as leads, perform follow-ups, analyze business activity, and use supervised machine learning to predict lead-conversion probability.

Proposed Project Title: SMARTLEAD — AI-Powered Digital Service Discovery & Lead Management System

Project Type: Full-Stack Multi-Domain Collaborative Student Project

Primary Technologies: Figma, HTML5, Tailwind CSS, React, Python, Django REST Framework, PostgreSQL, Scikit-learn

22. Practical Example — How SmartLead Works

The following simple example illustrates the complete journey of one user through the SmartLead platform. It is included so that students can understand the project by looking at realistic sample data rather than only reading feature descriptions.

22.1 Example Organization

Assume the platform is being developed for a fictional company called ABC Digital Solutions. The company provides Web Development, Mobile App Development, UI/UX Design, Digital Marketing and AI/ML services.

22.2 Sample User Credentials

These are demo credentials only. They illustrate the type of account and role information that may exist in the application.

User Type	Name	Email	Password (Demo)	Role
General User	Rahul Sen	rahul@example.com	Rahul@123	GENERAL_USER
Admin	Priya Das	admin@example.com	Admin@123	ADMIN
Super Admin	System Admin	superadmin@example.com	Super@123	SUPER_ADMIN

22.3 Sample Services

Service ID	Service	Category	Starting Price	Status
S001	Web Development	Software Development	₹40,000	Active
S002	Mobile App Development	Software Development	₹75,000	Active
S003	UI/UX Design	Design	₹25,000	Active
S004	Digital Marketing	Marketing	₹20,000/month	Active
S005	AI/ML Solutions	Artificial Intelligence	₹60,000	Active

22.4 Example User Journey

29. Rahul searches Google for “web development services”.
30. He reaches the SmartLead website through an SEO-optimized service page.
31. He opens the Web Development service page.
32. He reads the service description, features and starting price.
33. He clicks “Send Enquiry”.
34. He logs in or registers as a general user.
35. He submits his project requirement and budget.
36. The Django backend stores the enquiry in PostgreSQL.
37. The enquiry becomes a lead for administrative follow-up.
38. The ML component evaluates the lead and returns a conversion probability.
39. The admin sees the lead and prediction in the Admin Dashboard.
40. The admin contacts Rahul and changes the lead status as the sales process progresses.

22.5 Sample Enquiry Data

Field	Sample Value
Enquiry ID	ENQ10025
User	Rahul Sen
Service	Web Development
Traffic Source	Google Organic Search
Keyword	web development services
Pages Viewed	10
Time on Website	500 seconds
Previous Enquiries	2
Budget	₹50,000
Requirement	Responsive business website with admin panel
Enquiry Date	01-Oct-2026

22.6 Lead Created from the Enquiry

Lead Field	Sample Value
Lead ID	LEAD10025
Customer	Rahul Sen
Service	Web Development
Source	Google Organic Search
Status	NEW
Assigned To	Priya Das
Follow-up Date	03-Oct-2026

22.7 Machine Learning Prediction

The ML model receives selected historical and current lead features. For example:

Feature	Value	Model Input
Pages Viewed	10	10
Previous Enquiries	2	2
Time on Website	500 sec	500
Budget	₹50,000	50000
Engagement Score	85	85
Traffic Source	Google	Encoded value

Example model output: Conversion Probability = 81%

The dashboard may categorize this as a High Probability Lead. This is a prediction generated by the model and is not a guarantee that the lead will convert.

22.8 Admin Dashboard Example

Lead	Service	Prediction	Status
Rahul Sen	Web Development	81% — High	NEW
Ananya Roy	Digital Marketing	64% — Medium	CONTACTED
Sourav Paul	Mobile App Development	22% — Low	NOT INTERESTED

22.9 What Each Department Does in This Example

Department	What They Do	Example Contribution
Digital Marketing	Researches keywords and optimizes landing pages	Targets “web development services”
UI/UX	Designs the user and admin experience in Figma	Designs service page, enquiry form and lead dashboard
HTML/Tailwind	Creates responsive visual templates	Builds the service page and enquiry form
React	Makes the frontend functional	Displays services and submits enquiry

		through API
Django	Processes business logic and APIs	Stores ENQ10025 and creates/manages LEAD10025
PostgreSQL	Stores application data	Stores user, service, enquiry and lead records
Machine Learning	Predicts conversion probability	Returns 81% probability
Admin	Handles business follow-up	Contacts Rahul and updates lead status

Key understanding: The student is not simply building a website with unrelated pages. The example demonstrates one connected business process, from a marketing search through service discovery, enquiry, lead creation, ML prediction and administrative follow-up.